

How to set amount of hue levels in seaborn?

Asked 3 years, 2 months ago   Modified 3 years, 2 months ago   Viewed 9k times

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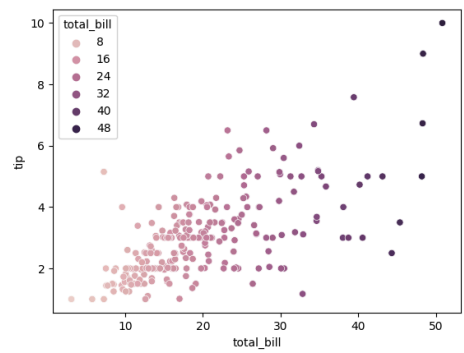
With the following code, I can use seaborn's scatterplot to plot data with certain colors assigned to data values.

How can I set the amount of colors that get used in this example? (e.g. if I want to have only two colors used or more than the 6 shown in the example)

```
import seaborn as sns
import matplotlib.pyplot as plt
tips = sns.load_dataset('tips')

print(tips.columns) # tips.columns= Index(['total_bill', 'tip',
'sex', 'smoker', 'day', 'time', 'size'], dtype=object)

sns.scatterplot(data=tips, x="total_bill", y="tip", hue="total_bill", )
```



python   seaborn   scatter-plot   Edit tags

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asked Dec 13, 2021 at 22:34

 7824238

410 ● 6 ● 15

- ▲
- Seaborn's `scatterplot` has a `legend=` keyword, which can be `'auto'`, `'brief'`, `'full'` or `False`. If you want something else, you'd need to create a custom legend. – JohanC Dec 13, 2021 at 22:40
- ▲
- Thanks for your suggestion. But shouldn't it perhaps be something around the hue parameter by itself that influences the amount of colors? Because if I omit the legend, still I would like to have e. g. two colors only. – 7824238 Dec 13, 2021 at 23:12
- ▲
- Sounds like a worthwhile [enhance request on seaborn](#), I encourage you to file. – smci Dec 13, 2021 at 23:58
- ▼

1 Answer

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Seaborn's `scatterplot` has a `legend=` keyword, which can be `'auto'`, `'brief'`, `'full'` or `False`. If you want something else, you'd need to create a custom legend.

If you want to control the exact number of levels and colors used, you can create an additional column with these hue levels. That way, you fully control the levels and their naming. Pandas `pd.cut()` can be helpful. It accepts either a number of bins, or a list of boundaries. You can set the names at the same time (the default labels would indicate the boundaries as `'[{start}, {end}]'`).

Here is an example:

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

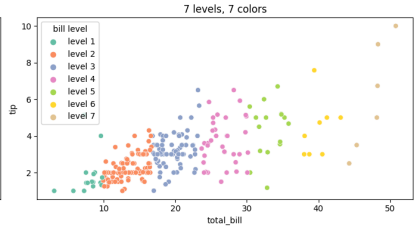
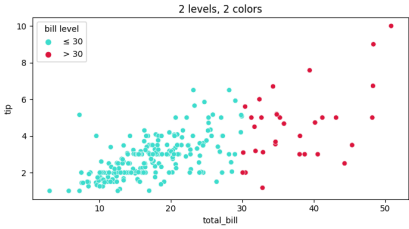
tips = sns.load_dataset('tips')

fig, (ax1, ax2) = plt.subplots(ncols=2, figsize=(14, 4))

tips['bill level'] = pd.cut(tips['total_bill'], [0, 30, 1000], labels=['≤ 30', '> 30'])
sns.scatterplot(data=tips, x="total_bill", y="tip", hue="bill level", palette=
['turquoise', 'crimson'], ax=ax1)
ax1.set_title('2 levels, 2 colors')

tips['bill level'] = pd.cut(tips['total_bill'], bins=7, labels=[f'level {i}' for
i in range(1, 8)])
sns.scatterplot(data=tips, x="total_bill", y="tip", hue="bill level",
palette='Set2', ax=ax2)
ax2.set_title('7 levels, 7 colors')

plt.tight_layout()
plt.show()
```



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edited Dec 13, 2021 at 23:57

answered Dec 13, 2021 at 23:50

 JohanC

80.4k ● 8 ● 50 ● 87

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- I'd also recommend looking into `hue_norm` and maybe passing a `boundarynorm`. It may be easier, although you won't get the nice customized legends you're producing here. – mwaskom Dec 14, 2021 at 1:19
- ▼